

CYBERSECURITY CASE STUDY

Suricata IDS/IPS Detection and Enforcement

OPNsense • Emerging Threats • Nmap • Network Monitoring

Alshaima Syed

Network Defense | Detection Engineering | SOC Analysis

Executive summary

This case study evaluates Suricata in both detection-only and prevention modes across a segmented OPNsense lab. Normal traffic established a clean baseline, after which controlled reconnaissance, service enumeration and ICMP abuse were generated from an external test host. The same activity was repeated with inline enforcement enabled, followed by an internal reconnaissance scenario to assess visibility into insider-originated behavior.

Outcome

Suricata distinguished normal activity from controlled attack traffic, generated scan and ICMP alerts with useful source/destination context, and actively disrupted selected traffic after IPS enforcement was enabled. Internal scanning also produced evidence, demonstrating that trusted-network activity still requires monitoring.

Environment	OPNsense firewall; Suricata; segmented WAN, LAN and DMZ; Kali test host; Metasploitable target.
Core tools	Suricata, Emerging Threats rules, OPNsense, Nmap and packet/firewall logs.
Security focus	Network visibility, reconnaissance detection, inline prevention and insider-threat monitoring.
Validation	Normal baseline followed by repeatable authorized attack simulations and evidence comparison.
Safety boundary	All testing occurred in an isolated lab against intentionally vulnerable systems.

Monitoring design

Suricata monitored traffic crossing the WAN, DMZ and LAN boundaries. Emerging Threats scan and ICMP categories supplied detection logic. The workflow separated observation from enforcement: IDS mode first documented what the rules detected, while IPS mode tested the operational effect of blocking.

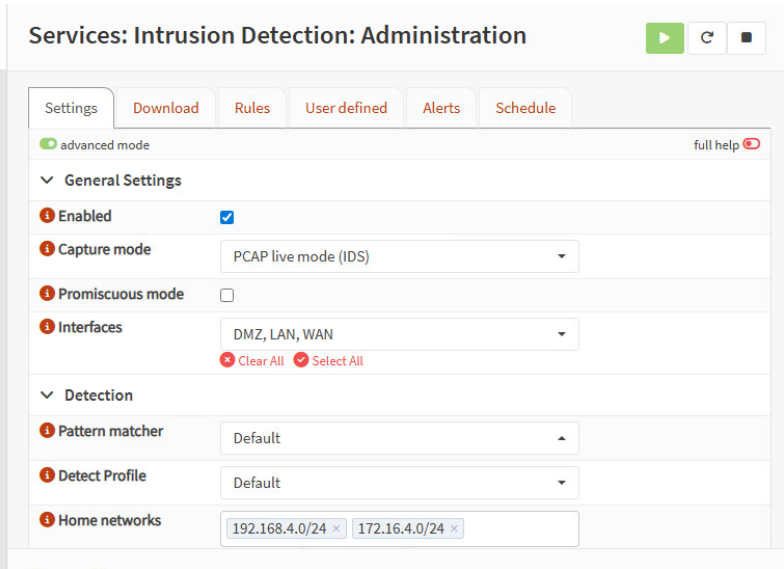


Figure 1. Suricata interface and monitoring configuration in OPNsense.

Phase 1: IDS detection

Baseline validation

- Generated ordinary HTTP, DNS and ICMP activity before attack simulation.
- Confirmed that routine browsing and connectivity testing did not create an unnecessary alert flood.
- Used the baseline to make later reconnaissance and rate-based activity easier to interpret.



Figure 2. Authorized DMZ target used for controlled detection testing.

Controlled reconnaissance

Host discovery, TCP port scanning and service-version detection were performed from the external test host against the DMZ target. Suricata recorded scan-related alerts and preserved the source, destination, interface and rule context needed for triage.

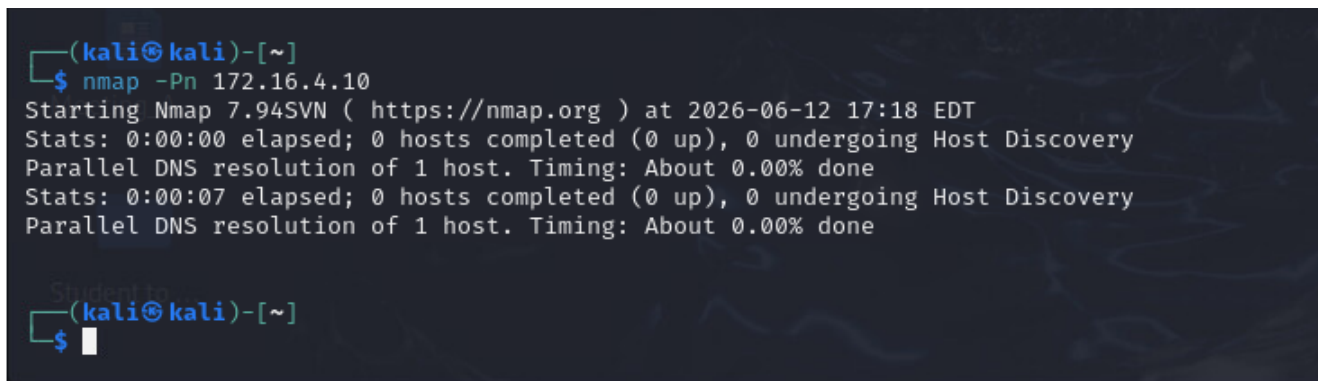
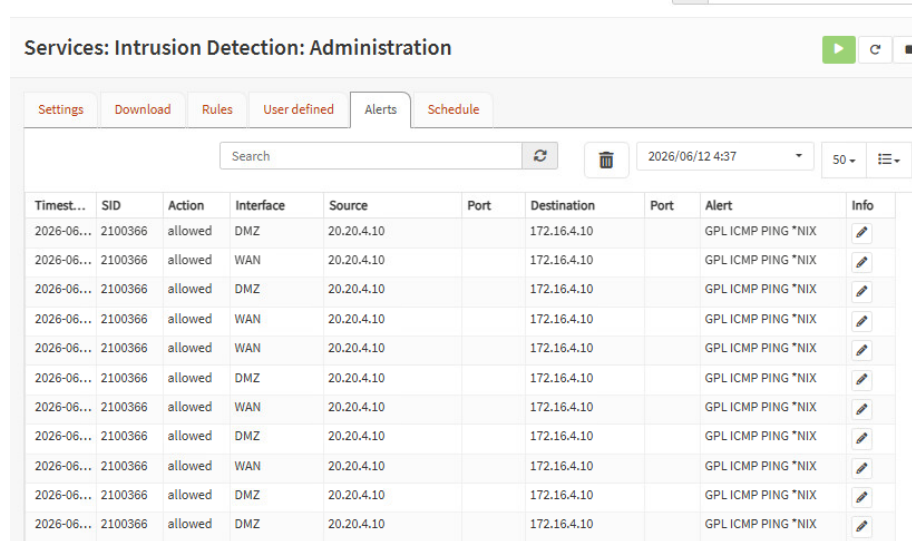


Figure 3. Controlled Nmap reconnaissance against the isolated DMZ target.

Alert analysis

The screenshot shows the 'Services: Intrusion Detection: Administration' interface. It has tabs for Settings, Download, Rules, User defined, Alerts, and Schedule. The Alerts tab is active, showing a table of alerts. The table has columns: Timest..., SID, Action, Interface, Source, Port, Destination, Port, Alert, and Info. There are 12 rows of alerts, all with SID 2100366 and Action 'allowed'. The Source is 20.20.4.10 and Destination is 172.16.4.10. The Alert text is 'GPL ICMP PING *NIX'.

Timest...	SID	Action	Interface	Source	Port	Destination	Port	Alert	Info
2026-06...	2100366	allowed	DMZ	20.20.4.10		172.16.4.10		GPL ICMP PING *NIX	
2026-06...	2100366	allowed	WAN	20.20.4.10		172.16.4.10		GPL ICMP PING *NIX	
2026-06...	2100366	allowed	DMZ	20.20.4.10		172.16.4.10		GPL ICMP PING *NIX	
2026-06...	2100366	allowed	WAN	20.20.4.10		172.16.4.10		GPL ICMP PING *NIX	
2026-06...	2100366	allowed	WAN	20.20.4.10		172.16.4.10		GPL ICMP PING *NIX	
2026-06...	2100366	allowed	DMZ	20.20.4.10		172.16.4.10		GPL ICMP PING *NIX	
2026-06...	2100366	allowed	WAN	20.20.4.10		172.16.4.10		GPL ICMP PING *NIX	
2026-06...	2100366	allowed	DMZ	20.20.4.10		172.16.4.10		GPL ICMP PING *NIX	
2026-06...	2100366	allowed	WAN	20.20.4.10		172.16.4.10		GPL ICMP PING *NIX	
2026-06...	2100366	allowed	DMZ	20.20.4.10		172.16.4.10		GPL ICMP PING *NIX	
2026-06...	2100366	allowed	DMZ	20.20.4.10		172.16.4.10		GPL ICMP PING *NIX	
2026-06...	2100366	allowed	DMZ	20.20.4.10		172.16.4.10		GPL ICMP PING *NIX	

Figure 4. Suricata alert list produced during controlled external activity.

Observed detection patterns

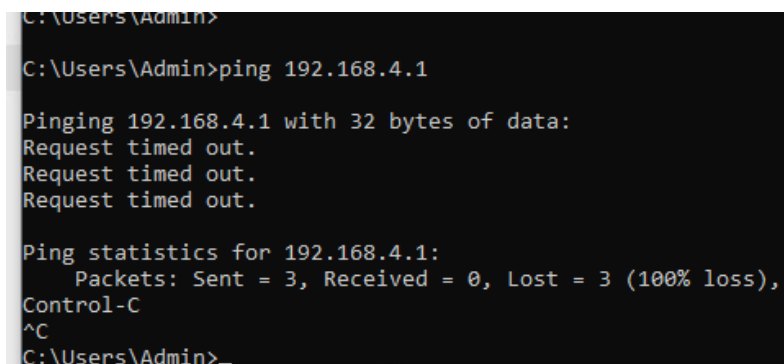
- Nmap activity triggered scan-oriented signatures associated with reconnaissance and service probing.
- ICMP traffic was logged with source and destination context; sustained volume produced repeated alerts consistent with abuse.
- Normal ICMP remained distinguishable from excessive traffic through rate, repetition and surrounding activity.
- Alert evidence was correlated with the exact test action rather than interpreted from signature wording alone.

Analyst interpretation

An alert indicates that observed traffic matched rule logic; it does not by itself prove malicious intent. Source role, target sensitivity, timing, frequency and authorized-scanner records must be considered before escalation.

Phase 2: IPS enforcement

Suricata was moved to inline prevention with blocking enabled while retaining the validated rule set. Repeating the host-discovery test produced destination-unreachable responses and complete packet loss in the observed attempt, providing direct evidence that enforcement altered the outcome.

The screenshot shows a Windows command prompt window. The user has entered 'ping 192.168.4.1'. The output shows three 'Request timed out.' messages. Below that, it shows 'Ping statistics for 192.168.4.1: Packets: Sent = 3, Received = 0, Lost = 3 (100% loss),'. The user has pressed 'Control-C' and '^C' to stop the command.

```
C:\Users\Admin>ping 192.168.4.1

Pinging 192.168.4.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.4.1:
    Packets: Sent = 3, Received = 0, Lost = 3 (100% loss),
Control-C
^C
C:\Users\Admin>
```

Figure 5. Repeated connectivity test after IPS enforcement, showing the blocked outcome.

IDS versus IPS

IDS mode	Detected and logged suspicious traffic while allowing the observed connection attempt to continue.
IPS mode	Inspected traffic inline and applied drop/block behavior to selected rule matches.
Operational benefit	IPS can reduce exposure by interrupting known unwanted patterns.
Operational risk	Poorly tuned inline rules can block legitimate traffic; staged validation and rollback planning are essential.

Services: Intrusion Detection: Administration

Settings Download Rules User defined Alerts Schedule

Search 2026/06/12 4:24 50

Time...	SID	Action	Interface	Source	Port	Destination	Port	Alert	Info
2026-06...	2010936	allowed	WAN	20.20.4.10	48610	172.16.4.10	1521	ET SCAN Suspicious in...	
2026-06...	2010939	allowed	WAN	20.20.4.10	48610	172.16.4.10	5432	ET SCAN Suspicious in...	
2026-06...	2010935	allowed	WAN	20.20.4.10	48610	172.16.4.10	1433	ET SCAN Suspicious in...	
2026-06...	2010937	allowed	WAN	20.20.4.10	48610	172.16.4.10	3306	ET SCAN Suspicious in...	
2026-06...	2010936	allowed	DMZ	20.20.4.10	48610	172.16.4.10	1521	ET SCAN Suspicious in...	
2026-06...	2002910	allowed	DMZ	20.20.4.10	48610	172.16.4.10	5801	ET SCAN Potential VN...	
2026-06...	2010939	allowed	DMZ	20.20.4.10	48610	172.16.4.10	5432	ET SCAN Suspicious in...	
2026-06...	2010935	allowed	DMZ	20.20.4.10	48610	172.16.4.10	1433	ET SCAN Suspicious in...	
2026-06...	2010937	allowed	DMZ	20.20.4.10	48610	172.16.4.10	3306	ET SCAN Suspicious in...	
2026-06...	2100366	allowed	DMZ	20.20.4.10		172.16.4.10		GPL ICMP PING *NIX	
2026-06...	2100366	allowed	WAN	20.20.4.10		172.16.4.10		GPL ICMP PING *NIX	

Showing 1 to 35 of 35 entries

Figure 6. Enforcement-stage Suricata alerts supporting the IDS/IPS comparison.

Phase 3: internal reconnaissance

A controlled internal host performed subnet discovery and a port scan against an internal Windows system. The activity demonstrated how a legitimate network position can be misused for discovery and lateral-movement preparation. Suricata evidence confirmed that monitoring cannot stop at the internet perimeter.

```

kali@kali:~$ nmap -sS -p 172.16.4.10
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-12 00:40 EDT
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host: Timing: About 0.00% done
Stats: 0:00:12 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 2 hosts: Timing: About 0.00% done
Stats: 0:00:22 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan timing: About 21.74% done; ETC: 00:41 (0:00:22 remaining)
Nmap scan report for 172.16.4.10
Host is up (0.00042s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  netbios-ssn
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  java-rmi
1524/tcp  open  bindshell
2049/tcp  open  nfs
2121/tcp  open  cproxy-ftpt
3386/tcp  open  mysql
5432/tcp  open  postgresql
5980/tcp  open  vnc
6080/tcp  open  X11
6667/tcp  open  irc
8080/tcp  open  ajp13
8180/tcp  open  http
Service Info: Hosts: metasploitable.localdomain, irc.metasploitable.LAN; OS: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 189.18 seconds
kali@kali:~$

```

Figure 7. Authorized internal reconnaissance used to test insider-threat visibility.

Security analysis and recommendations

Key findings

- Segment-boundary monitoring provided useful visibility into both external and internal reconnaissance.
- Detection-only deployment supported tuning and baseline learning before enforcement.
- Inline prevention changed the observed result, but block evidence still required correlation across alerts and connectivity tests.
- Internal trust should be limited; east-west telemetry is important for identifying discovery and lateral-movement precursors.

Production improvements

- Begin with IDS baselining, tune noisy signatures, then move selected high-confidence rules to IPS with change control.
- Centralize Suricata EVE JSON, firewall and endpoint telemetry in a SIEM for correlation and retention.
- Maintain an allowlist of approved vulnerability scanners and suppress only narrowly defined known-good activity.
- Measure false-positive rate, detection coverage, mean triage time and blocked legitimate sessions.
- Protect management interfaces, update signatures through a controlled process and test rollback procedures.
- Map validated behaviors to MITRE ATT&CK; only after confirming the observed technique and context.

Skills demonstrated

IDS/IPS engineering	Suricata deployment, interface selection, rule categories and staged enforcement.
Network analysis	Baseline comparison, source/destination correlation and segmented traffic interpretation.
Detection validation	Repeatable attack simulation, alert review and outcome-based verification.
SOC judgment	Distinguishing signature matches from confirmed malicious activity.
Insider-threat monitoring	Visibility into trusted-zone reconnaissance and lateral-movement precursors.

Conclusion

The project demonstrated a disciplined progression from visibility to prevention. Baseline traffic established context, IDS mode showed what Suricata could detect, IPS mode demonstrated selected enforcement, and the internal scenario reinforced the need for monitoring beyond the perimeter.

Selected references

Suricata User Guide — <https://docs.suricata.io/>
OPNsense Intrusion Prevention System — <https://docs.opnsense.org/manual/ips.html>
Emerging Threats Open Rules — <https://rules.emergingthreats.net/>
MITRE ATT&CK Network Service Discovery — <https://attack.mitre.org/techniques/T1046/>